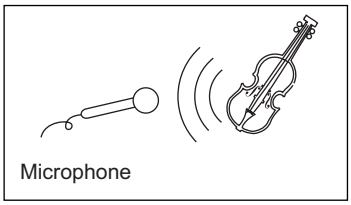
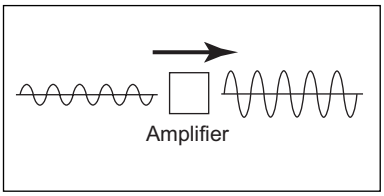
Sending and receiving music by radio

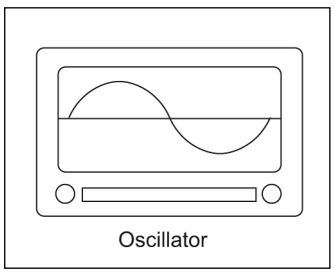
8a

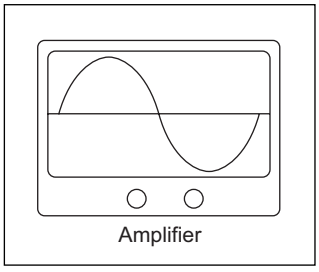
Studio band music is collected by microphones, manipulated and eventually the changed signal is transmitted from the radio station's antenna. Far away (or near by), a radio receiver picks up those radio waves and uses them to reproduce the studio music. Some details of the steps in the process are described below.

1 Match the writing about a radio transmitter to the ordered diagrams by drawing lines to connect them.









A
An oscillator produces high, constant frequency electric vibrations called a carrier wave. For example, ABC Sydney on the AM band has a carrier wave frequency of 702 kHz.

B
Sound waves are changed into varying low frequency electric impulses by a microphone. The frequency and strength of the impulses depend on the pitch (i.e. frequency) and loudness (amplitude) of the sounds.

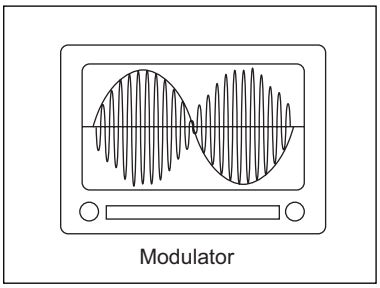
C
The audio wave and carrier wave are merged into a high voltage modulated wave by a modulator.

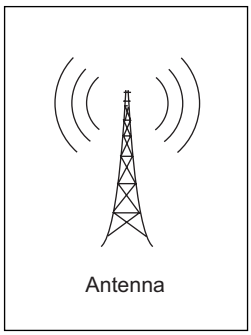
D
The carrier wave is amplified by an intermediate amplifier.

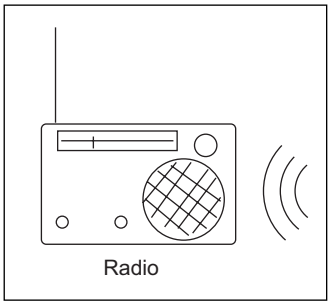
E
The electric impulses, forming the audio wave, are strengthened by an amplifier.

F
The high voltage, high frequency modulated waves produce radio waves through an antenna. The waves travel at 300 000 km/s away from the antenna.

G
The radio station JJJ on the FM band has a carrier wave frequency of 105.7 MHz in Sydney. It is these waves that are collected by a radio and changed into sound.







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Sending and receiving music by radio

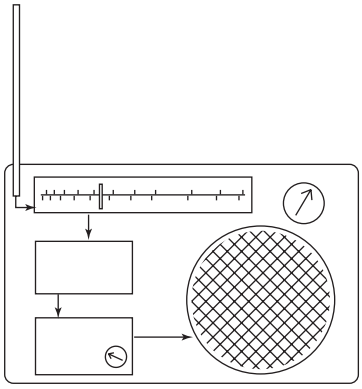
8b

2 Refer to the previous page. Write an explanation for how a radio station changes music into radio waves. Do not forget to write introductory and concluding paragraphs. Using time and cause conjunctions such as when, after, then, because, since and therefore will help the clarity of your explanation.

3 The sentences below describe the operation of a radio receiver. Use the sentence letters to label the diagram showing the part that performs the operation described in the sentence.

Letters	Sentences	Order
A	An alternating current of a particular frequency is selected by the **tuner** and passed on.	
B	In the **demodulator** the carrier wave is separated from the signal it was carrying. Only the audio wave is passed on.	
C	Radio waves are picked up by the **antenna**, which generates weak alternating currents.	
D	The amplified audio wave is converted into sound and music by the **speaker**.	
E	The audio wave is amplified to a desired loudness by the **amplifier**.	

Radio receiver



4 Order the sentences (starting with the antenna) to show the change from radio wave to sound wave by writing 1, 2, etc. in the Order column of the table above.

5 Label the diagram of the radio above using the bold terms in the sentences.

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Sending and receiving music by radio

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6 The sentences in question 3 begin with 'waves' or 'current' because the focus is on waves and currents. The sentences are all in passive voice. Change them to active voice by beginning the sentences with the bold terms. Write the sentences in your numbered order.

7 Use information from the drawing and the sentences to write an explanation of how a radio produces music from radio waves. Draw and label a diagram in the space below to illustrate your explanation.

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